Lecture 10: Data mining and simple linear regression

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22 October 2026

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1 What this document is

This is Lecture 10 as a written tutorial, in the same order as the script `lecture10_regression.R`. It follows Chapter 8 of *Business Analytics* (Camm et al.) and Chapter 5.7 of *Tidyverse Skills for Data Science* (Wright et al.). Read it after the lecture, or run it: open the `.Rmd` file in RStudio, press **Knit**, and this PDF is rebuilt on your computer.

In Lecture 7 you measured how strongly two variables move together with the correlation. In Lectures 8 and 9 you drew them with `ggplot2`. Today you build your first model: a straight line that predicts one variable from another.

2 Packages and the data

Two packages. Install each once per computer; running the install again does no harm.

```
install.packages("tidyverse")
install.packages("broom")
```

Load them every session:

```
library(tidyverse)
library(broom)
data(trees)
glimpse(trees)
```

```
#> Rows: 31
#> Columns: 3
#> $ Girth  <dbl> 8.3, 8.6, 8.8, 10.5, 10.7, 10.8, 11.0, 11.0, 11.1, 11.2, 11.3, ~
#> $ Height <dbl> 70, 65, 63, 72, 81, 83, 66, 75, 80, 75, 79, 76, 76, 69, 75, 74,~
#> $ Volume <dbl> 10.3, 10.3, 10.2, 16.4, 18.8, 19.7, 15.6, 18.2, 22.6, 19.9, 24.~
```

`trees` comes with R itself. It describes 31 black cherry trees:

- `Girth` is the trunk **diameter** in inches, measured 4.5 feet above the ground. The name is misleading; R's help page `?trees` says it is the diameter.
- `Height` is the height of the tree in feet.
- `Volume` is the volume of timber in cubic feet.

One inch is 2.54 cm and one foot is 0.30 m.

3 Data mining

Data mining is the use of software to find useful patterns in large amounts of data. It comes in two kinds:

- **Descriptive data mining**, also called *unsupervised learning*: there is no answer y to predict. The aim is to find groups and patterns. Example: a shop groups its customers by what they buy.
- **Predictive data mining**, also called *supervised learning*: learn from past cases where the answer y is known, then predict y for new cases. Example: predict the price of a house from its size. When y is a number, the task is called **regression**; when y is a category, **classification**.

The variable we predict, y, is the *dependent variable*, *outcome* or *target*. The variable we predict it from, x, is the *independent variable*, *predictor* or *feature*. **Simple linear regression** uses one x and a straight line.

3.1 The model

$$\mathbf{y} = \alpha + \beta\mathbf{x} + \varepsilon$$

- α, the **intercept**: the value of y when $x = 0$, where the line crosses the y-axis.
- β, the **slope**: rise over run, the change in y divided by the change in x. When x goes up by 1, y goes up by β on average. A slope of 0 means no linear relationship.
- ε, the **error term**: everything else that affects y, which the line does not capture. For a tree: age, soil, light.

Example: consumption $= \alpha + \beta\times$ income $+\varepsilon$. Here β is how much consumption rises when income rises by one unit.

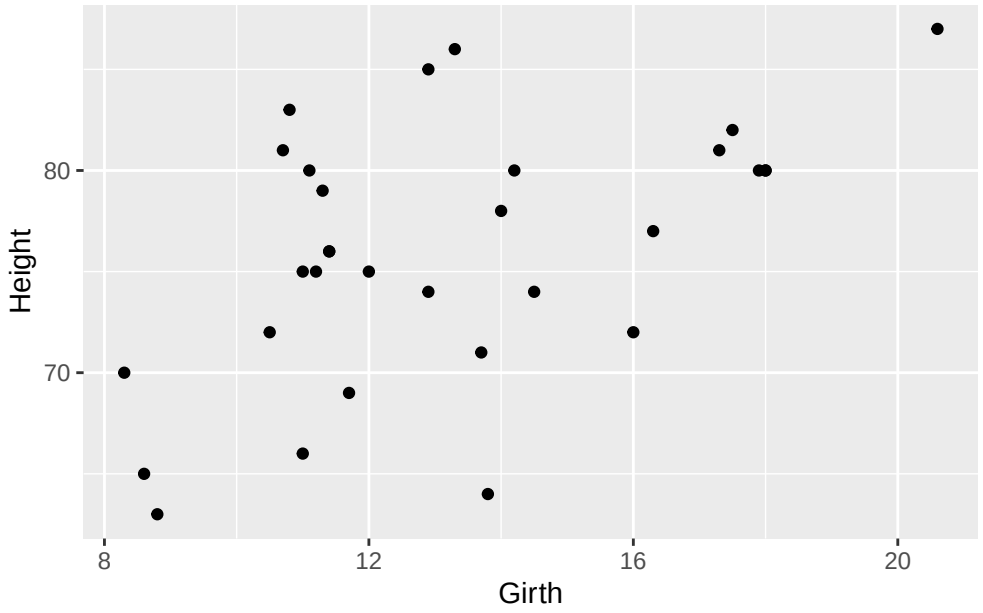
α and β are the true values for the whole population, which we never know. From a sample we estimate them; the estimates are written $\hat{\alpha}$ and $\hat{\beta}$ ("alpha-hat", "beta-hat"), and the fitted line is $\hat{y} = \hat{\alpha} + \hat{\beta}x$.

4 A first look at two variables

The question for today: the diameter of a tree is easy to measure, but the height of a standing tree is not. Can we predict a tree's height from its diameter? So y is `Height` and x is `Girth`.

Always look at the data before you fit anything:

```
ggplot(trees, aes(x = Girth, y = Height)) +
  geom_point()
```




```
cor(trees$Girth, trees$Height)
```

```
#> [1] 0.5192801
```

The predictor goes on the horizontal axis and the outcome on the vertical axis. The points go up: thicker trees tend to be taller, but with much scatter. The correlation, $r = 0.52$, is positive and moderate.

5 Fitting the line

5.1 Least squares

A **residual** is actual minus fitted, $e = y - \hat{y}$: how far a point lies above or below the line. The **least squares method** picks the line with the smallest sum of squared residuals. The residuals are squared so that the ones above the line (positive) and below it (negative) do not cancel.

5.2 Fitting it in R

`lm()` fits a linear model. Read the formula `Height ~ Girth` as "height explained by girth": the outcome on the left of the tilde, the predictor on the right. The general form is `lm(y ~ x, data = table)`.

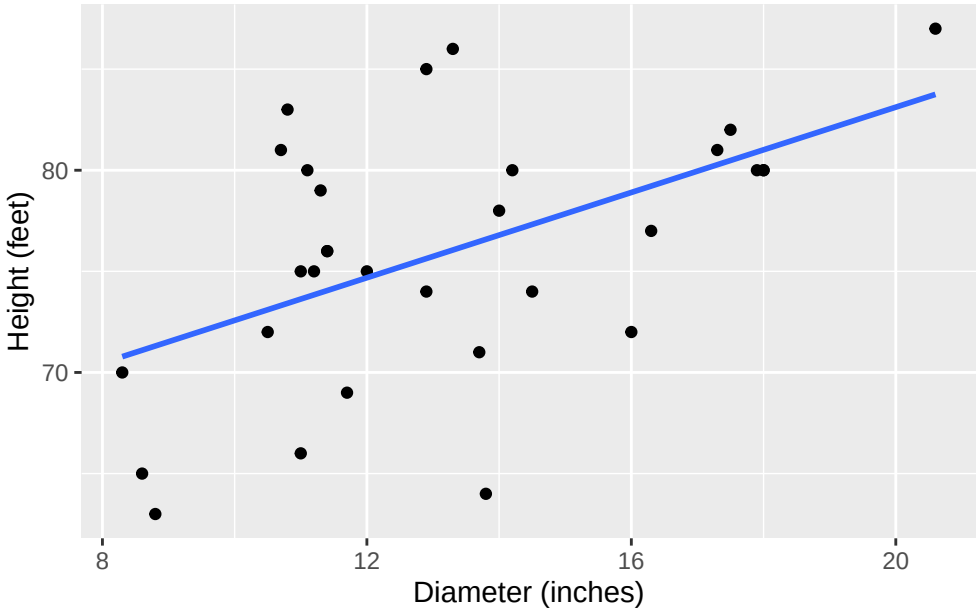
```
fit <- lm(Height ~ Girth, data = trees)
fit
```

```
#>
#> Call:
#> lm(formula = Height ~ Girth, data = trees)
#>
#> Coefficients:
#> (Intercept)        Girth
#>      62.031        1.054
```

The estimates replace α and β: **Height = 62.03 + 1.054 × Girth**.

- **Slope** $\hat{\beta} = 1.054$**.** One inch more diameter goes with 1.05 feet more height, on average. The unit of the slope is feet per inch.
- **Intercept** $\hat{\alpha} = 62.03$**.** The height the line gives for a diameter of 0. No tree is that thin (the diameters run from 8.3 to 20.6 inches), so it has no real meaning; it only positions the line.

```
ggplot(trees, aes(x = Girth, y = Height)) +
  geom_point() +
  geom_smooth(method = "lm", se = FALSE) +
  labs(x = "Diameter (inches)", y = "Height (feet)")
```



`geom_smooth(method = "lm")` runs the same `lm()` and draws the result. `se = FALSE` draws the line only. In RStudio you will also see the message `geom_smooth() using formula = 'y ~ x'`: that is information, not an error.

5.3 Fitted values, residuals and predictions

The **fitted value** of a tree is the point on the line at its diameter: the height the line says that tree should have. The residual is how far the real height is from it.

```
trees_fit <- trees |>
  mutate(fitted   = fitted(fit),
         residual = residuals(fit))
head(trees_fit, 3)
```

```
#>   Girth Height Volume   fitted  residual
#> 1   8.3     70   10.3 70.78257 -0.782575
#> 2   8.6     65   10.3 71.09889 -6.098886
#> 3   8.8     63   10.2 71.30976 -8.309759
```

Tree 2 is 65 feet tall, but the line says 71.1 feet, so its residual is −6.1: it lies 6.1 feet below the line.

`predict()` gives the line's answer for a new tree:

```
predict(fit, newdata = tibble(Girth = 15))
```

```
#>        1
#> 77.84685
```

A tree 15 inches across is predicted to be 77.8 feet tall. Predict only inside the range of the data, here 8.3 to 20.6 inches. Outside that range is **extrapolation**: the data cannot show whether the line still holds there.

6 How sure are we about the estimates?

The 31 trees are a sample. Another 31 black cherry trees would give slightly different estimates. `summary()` reports how much they could change.

```
summary(fit)
```

```
#>
#> Call:
#> lm(formula = Height ~ Girth, data = trees)
#>
#> Residuals:
#>      Min       1Q   Median       3Q      Max
#> -12.5816  -2.7686   0.3163   2.4728   9.9456
#>
#> Coefficients:
#>             Estimate Std. Error t value Pr(>|t|)
#> (Intercept)  62.0313     4.3833  14.152 1.49e-14 ***
#> Girth         1.0544     0.3222   3.272  0.00276 **
#> ---
#> Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
#>
#> Residual standard error: 5.538 on 29 degrees of freedom
#> Multiple R-squared:  0.2697, Adjusted R-squared:  0.2445
#> F-statistic: 10.71 on 1 and 29 DF,  p-value: 0.002758
```

In the `Coefficients` table, the row `Girth` holds four numbers about the slope, and the row `(Intercept)` the same four about the intercept:

- **Estimate, 1.0544**: the slope, the beta estimate.
- **Std. Error, 0.3222**: the **standard error**: how much the estimate would change from one sample to the next. A small standard error means a precise estimate. It gets smaller with more data and with points closer to the line.

- **t value, 3.272**: the estimate divided by its standard error. The slope lies 3.3 standard errors away from 0.
- **Pr(>|t|), 0.00276**: the **p-value**.

To read the p-value, suppose diameter and height had no linear relationship at all, so that the true slope were 0. The p-value is the chance of getting a slope at least this far from 0 in that case. Here it is 0.0028: fewer than 3 samples in 1,000.

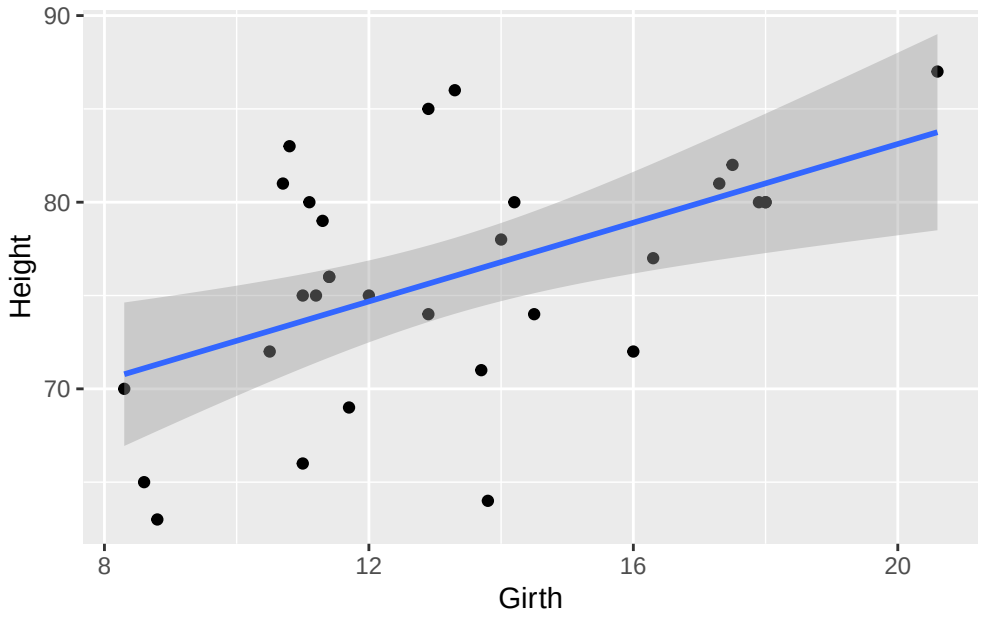
Decision rule: a p-value below 0.05 makes the estimate **statistically significant**: we conclude that the true slope is not 0, so diameter does tell us something about height. The stars next to it say the same thing (** means p is between 0.001 and 0.01). The p-value does not say how big or how useful the slope is, and it is not the chance that our slope is wrong.

With the estimates, their standard errors and p-values, the fitted model reads:

Height = 62.03 + 1.054 × Girth, with standard errors (4.38) and (0.32), and p-values $p < 0.001$ and $p = 0.003$.

The same uncertainty can be drawn as a band around the line, the default of `geom_smooth()`:

```
ggplot(trees, aes(x = Girth, y = Height)) +
  geom_point() +
  geom_smooth(method = "lm")
```



The grey band shows where the true line plausibly runs. It is narrowest at the average diameter and widens towards the ends.

7 How good is the line?

7.1 R-squared

R-squared is the share of the variation in y that the line explains, from 0 (nothing) to 1 (every point on the line). `summary(fit)` reports `Multiple R-squared: 0.2697`: 27% of the differences in height

between the trees go with differences in diameter. The remaining 73% is unexplained: age, soil, light and other things, the error term. With one x, R-squared equals the correlation squared:

```
cor(trees$Girth, trees$Height)^2
```

```
#> [1] 0.2696518
```

The **residual standard error**, 5.538 feet, is the typical distance of a tree from the line. It is not the same as the standard error of the slope.

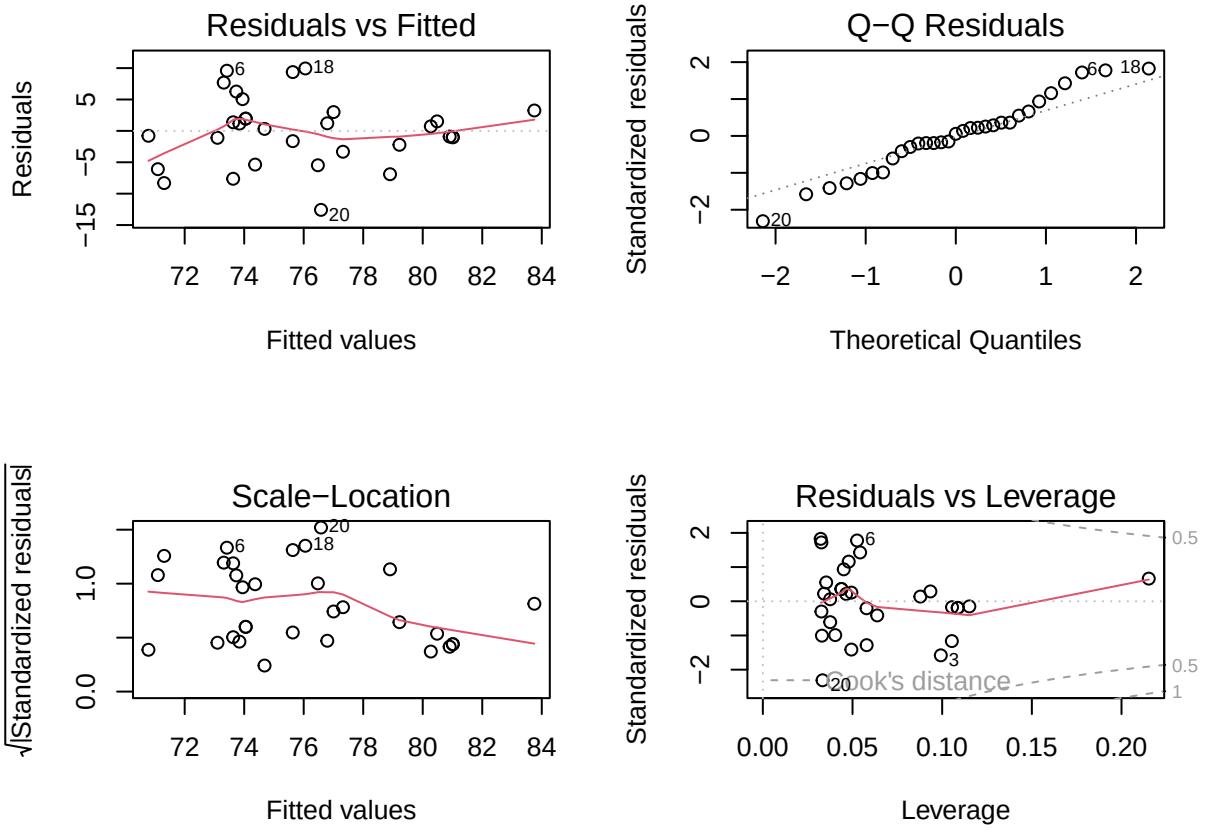
7.2 The error term and the four conditions

The error term collects everything that affects y apart from x. The standard error and the p-value are reliable when:

1. **Linearity**: the points follow a straight line, not a curve.
2. **Independence**: one observation does not affect another. Each tree is measured once.
3. **Equal spread**: the points are about as far from the line everywhere; no funnel.
4. **Normal residuals**: most residuals are small, few are large, roughly symmetric.

R draws four check plots for a fitted model:

```
par(mfrow = c(2, 2))
plot(fit)
```



```
par(mfrow = c(1, 1))
```

`par(mfrow = c(2, 2))` puts the four plots in one window, 2 rows by 2 columns; the last line sets it back. Look for clear patterns; with 31 trees the red lines always wiggle a little.

- **Residuals vs Fitted**: look for a U or an arch. Here there is none: linearity holds.
- **Q-Q Residuals**: the points lie near the dotted line, so the residuals are roughly normal.
- **Scale-Location**: look for a clear rise or fall. Here the red line falls a little on the right. With 31 trees that is mild, so equal spread roughly holds.
- **Residuals vs Leverage**: no point lies beyond the dashed Cook's distance lines, so no single tree drives the line.

7.3 The results as tables

The `broom` package gives the results of a model as tables (tibbles): `tidy()` for the coefficients, `glance()` for one row about the whole model, and `augment()` for one row per observation.

```
tidy(fit)
```

```
#> # A tibble: 2 x 5
#>   term        estimate std.error statistic    p.value
#>   <chr>          <dbl>     <dbl>     <dbl>      <dbl>
#> 1 (Intercept)  62.031    4.3833    14.152  1.4930e-14
#> 2 Girth         1.0544   0.32222    3.2722 2.7578e- 3
```

```
glance(fit)
```

```
#> # A tibble: 1 x 12
#>   r.squared adj.r.squared  sigma statistic   p.value    df  logLik    AIC    BIC
#>       <dbl>         <dbl>  <dbl>     <dbl>     <dbl> <dbl>   <dbl>  <dbl>  <dbl>
#> 1   0.26965       0.24447 5.5385    10.707 0.0027578     1 -96.017 198.03 202.34
#> # i 3 more variables: deviance <dbl>, df.residual <int>, nobs <int>
```

```
head(augment(fit), 3)
```

```
#> # A tibble: 3 x 8
#>   Height Girth .fitted   .resid     .hat .sigma   .cooksd .std.resid
#>    <dbl> <dbl>   <dbl>    <dbl>    <dbl>  <dbl>     <dbl>      <dbl>
#> 1     70   8.3  70.783 -0.78257 0.11514  5.6343 0.0014680   -0.15021
#> 2     65   8.6  71.099 -6.0989  0.10540  5.5032 0.079846    -1.1642
#> 3     63   8.8  71.310 -8.3098  0.099237 5.3882 0.13766     -1.5809
```

8 Correlation and causation

A regression measures how x and y move together. On its own it says nothing about cause and effect.

- **A confounder** is a third variable that drives both x and y. Thicker trees tend to be taller, but a thicker trunk does not make a tree grow taller: older trees are both thicker and taller, so age drives both. Likewise, ice cream sales and sunburn cases rise and fall together because hot, sunny days drive both.
- **Reverse causation**: y may drive x.
- **Coincidence**: among many variables, some pairs correlate by chance.

A cause is shown by a controlled experiment, where only x is changed on purpose, not by a regression on observed data.

9 What you can do now

- say what predictive and descriptive data mining are
- fit a simple linear regression with `lm()` and draw it with `geom_smooth()`
- read the intercept and the slope in the units of the data, and predict
- read the standard error and the p-value, and apply the decision rule
- read R-squared, and check the conditions with the check plots
- explain why a strong relationship does not prove cause and effect

10 Sources

- Camm, J. D., Cochran, J. J., Fry, M. J. and Ohlmann, J. W. (2024). *Business Analytics* (5th ed.). Cengage. Ch. 1.3 and the introductions to Ch. 6 and Ch. 10 (data mining); Ch. 8 *Linear Regression*, sections 8.1 to 8.3 and 8.5.
- Wright, C., Ellis, S., Hicks, S. and Peng, R. (2021). *Tidyverse Skills for Data Science in R*, Ch. 5.7 *Linear Modeling*. https://jhudatascience.org/tidyversecourse/model.html
- The `trees` data: Ryan, T. A., Joiner, B. L. and Ryan, B. F. (1976). *The Minitab Student Handbook*. Duxbury. In R: `?trees`.
- Course repository: https://github.com/touseefhameed/oka2015-applied-data-science